

Table 1: Performance of DetectGPT and Fast-DetectGPT concerning TPR@FPR-0.5%/100. The detection model is trained on ChatGPT text.

Dataset	Method	ChatGPT	GPT-4	ChatGPT-turbo	ChatGLM	Dolly	Claude	avg
Essay	DetectGPT	37.04 $\pm$ 10.21	24.75 $\pm$ 10.03	5.52 $\pm$ 1.93	21.45 $\pm$ 14.25	15.45 $\pm$ 7.05	4.48 $\pm$ 2.57	18.12
	Fast-DetectGPT	9.92 $\pm$ 11.83	37.38 $\pm$ 23.92	51.52 $\pm$ 24.00	15.66 $\pm$ 24.11	0.09 $\pm$ 0.17	0.96 $\pm$ 1.55	19.25
Reuters	DetectGPT	4.40 $\pm$ 2.62	0.64 $\pm$ 0.54	2.32 $\pm$ 1.87	2.56 $\pm$ 2.80	0.48 $\pm$ 0.47	3.04 $\pm$ 1.61	2.24
	Fast-DetectGPT	72.00 $\pm$ 5.47	9.28 $\pm$ 2.44	94.08 $\pm$ 0.82	93.04 $\pm$ 2.10	1.20 $\pm$ 0.91	9.04 $\pm$ 3.23	46.44
SQuAD1	DetectGPT	0.35 $\pm$ 0.69	0.36 $\pm$ 0.48	0.47 $\pm$ 0.44	0.58 $\pm$ 0.73	0.58 $\pm$ 0.37	0.81 $\pm$ 0.28	0.52
	Fast-DetectGPT	2.66 $\pm$ 2.12	3.45 $\pm$ 1.97	0.94 $\pm$ 0.47	1.73 $\pm$ 1.21	2.67 $\pm$ 1.14	18.27 $\pm$ 7.39	4.95

Table 2: Performance of DetectGPT and Fast-DetectGPT concerning AUROC/100. The detection model is trained on ChatGPT text.

Dataset	Method	ChatGPT	GPT-4	ChatGPT-turbo	ChatGLM	Dolly	Claude	avg
Essay	DetectGPT	96.86 $\pm$ 0.97	95.64 $\pm$ 0.45	44.40 $\pm$ 1.73	95.60 $\pm$ 0.62	92.55 $\pm$ 0.58	46.36 $\pm$ 0.71	78.57
	Fast-DetectGPT	96.41 $\pm$ 0.96	94.82 $\pm$ 1.19	99.20 $\pm$ 0.22	97.91 $\pm$ 0.49	84.16 $\pm$ 1.97	89.19 $\pm$ 0.76	93.61
Reuters	DetectGPT	92.78 $\pm$ 1.34	85.71 $\pm$ 2.02	49.24 $\pm$ 1.65	91.59 $\pm$ 1.67	83.41 $\pm$ 1.73	66.59 $\pm$ 2.53	78.22
	Fast-DetectGPT	97.70 $\pm$ 0.16	70.76 $\pm$ 1.59	99.11 $\pm$ 0.12	99.05 $\pm$ 0.20	51.26 $\pm$ 1.29	83.49 $\pm$ 1.70	83.56
SQuAD1	DetectGPT	52.60 $\pm$ 2.74	47.47 $\pm$ 1.42	52.05 $\pm$ 1.87	50.88 $\pm$ 2.47	50.24 $\pm$ 2.20	51.12 $\pm$ 2.44	50.73
	Fast-DetectGPT	65.04 $\pm$ 0.10	63.47 $\pm$ 1.23	62.18 $\pm$ 0.40	61.24 $\pm$ 0.61	63.64 $\pm$ 0.78	77.74 $\pm$ 0.55	65.55

Table 3: Performance comparison (TPR@FPR-0.5%/100) in DetectRL dataset. The detection model is trained on Google-PaLM text.

Method	Google-PaLM	Claude-instant	ChatGPT	Llama-2-70b	Avg.
ChatGPT-D	87.37 $\pm$ 8.94	67.58 $\pm$ 11.18	69.64 $\pm$ 13.03	99.12 $\pm$ 0.68	80.93
ChatGPT-STK	90.11$\pm$4.85	71.73$\pm$7.41	73.56$\pm$12.67	99.21$\pm$0.51	83.65
OpenAI-D	99.00$\pm$0.26	68.59 $\pm$ 2.56	98.94 $\pm$ 0.64	99.91$\pm$0.09	91.61
OpenAI-STK	98.98 $\pm$ 0.28	70.96$\pm$3.15	99.08$\pm$0.66	99.91$\pm$0.09	92.23
MPU	96.39 $\pm$ 1.43	86.51 $\pm$ 1.70	99.38 $\pm$ 0.73	99.87 $\pm$ 0.08	95.54
MPU-STK	96.41$\pm$1.33	88.41$\pm$0.85	99.41$\pm$0.59	99.91$\pm$0.08	96.03

Table 4: Performance comparison (AUROC/100) in DetectRL dataset. The detection model is trained on Google-PaLM text.

Method	Google-PaLM	Claude-instant	ChatGPT	Llama-2-70b	Avg.
ChatGPT-D	99.06 $\pm$ 0.43	96.72 $\pm$ 1.39	99.31 $\pm$ 0.42	99.94$\pm$0.04	98.76
ChatGPT-STK	99.18$\pm$0.32	96.89$\pm$0.92	99.48$\pm$0.24	99.94$\pm$0.04	98.87
OpenAI-D	99.91$\pm$0.02	95.57 $\pm$ 0.63	99.79 $\pm$ 0.05	100.00$\pm$0.01	98.82
OpenAI-STK	99.91$\pm$0.02	96.20$\pm$0.76	99.81$\pm$0.08	100.00$\pm$0.01	98.98
MPU	99.81$\pm$0.08	99.24 $\pm$ 0.11	99.94$\pm$0.04	100.00$\pm$0.00	99.74
MPU-STK	99.80 $\pm$ 0.07	99.32$\pm$0.11	99.94$\pm$0.04	100.00$\pm$0.00	99.77

Table 5: Performance comparison (TPR@FPR-0.5%/100) against paraphrase attack Polishing. The detection model is trained on paraphrased Google-PaLM text.

Method	Google-PaLM	Claude-instant	ChatGPT	Llama-2-70b	avg.
ChatGPT-D	85.73 $\pm$ 7.41	70.11 $\pm$ 10.93	94.90 $\pm$ 3.18	97.66 $\pm$ 1.74	87.10
ChatGPT-STK	87.95 $\pm$ 7.85	76.33 $\pm$ 15.11	92.29 $\pm$ 8.72	97.67 $\pm$ 2.94	88.56
OpenAI-D	99.70 $\pm$ 0.18	89.59 $\pm$ 1.95	99.98 $\pm$ 0.03	100.00 $\pm$ 0.00	97.32
OpenAI-STK	99.73 $\pm$ 0.19	89.92 $\pm$ 1.64	99.98 $\pm$ 0.03	100.00 $\pm$ 0.00	97.41
MPU	98.60 $\pm$ 0.27	95.26 $\pm$ 1.10	99.97 $\pm$ 0.04	100.00 $\pm$ 0.00	98.46
MPU-STK	98.52 $\pm$ 0.21	95.46 $\pm$ 1.13	99.97 $\pm$ 0.04	100.00 $\pm$ 0.00	98.49

Table 6: Performance comparison (TPR@FPR-0.5%/100) against paraphrase attack Back-translation. The detection model is trained on paraphrased Google-PaLM text.

Method	Google-PaLM	Claude-instant	ChatGPT	Llama-2-70b	Avg.
ChatGPT-D	87.88 $\pm$ 6.59	85.35 $\pm$ 6.26	95.90 $\pm$ 3.39	97.21 $\pm$ 2.21	91.58
ChatGPT-STK	90.06 $\pm$ 2.35	87.22 $\pm$ 2.71	96.67 $\pm$ 2.03	97.85 $\pm$ 1.03	92.95
OpenAI-D	99.64 $\pm$ 0.08	99.58 $\pm$ 0.18	99.48 $\pm$ 0.15	99.40 $\pm$ 0.29	99.52
OpenAI-STK	99.62 $\pm$ 0.13	99.59 $\pm$ 0.17	99.50 $\pm$ 0.21	99.40 $\pm$ 0.25	99.53
MPU	95.90 $\pm$ 0.71	94.25 $\pm$ 1.19	99.78 $\pm$ 0.12	99.78 $\pm$ 0.13	97.43
MPU-STK	95.97 $\pm$ 0.79	94.35 $\pm$ 1.25	99.78 $\pm$ 0.12	99.78 $\pm$ 0.13	97.47

Table 7: Cross-domain performance concerning TPR@FPR-0.5%/100. The detection model is trained on ChatGPT text, with the source domain as Essay and the target domain as Reuters.

Method	ChatGPT	GPT-4	ChatGPT-turbo	ChatGLM	Dolly	Claude	Avg.
ChatGPT-D	90.64 $\pm$ 3.07	83.84 $\pm$ 2.97	84.40 $\pm$ 9.06	97.28 $\pm$ 0.64	39.76 $\pm$ 5.36	2.48 $\pm$ 1.48	66.40
ChatGPT-STK	94.48 $\pm$ 1.80	86.08 $\pm$ 1.22	87.52 $\pm$ 4.90	98.00 $\pm$ 0.84	45.28 $\pm$ 4.75	10.00 $\pm$ 3.23	70.23
OpenAI-D	79.52 $\pm$ 35.36	68.40 $\pm$ 20.47	78.96 $\pm$ 33.48	83.60 $\pm$ 28.81	37.52 $\pm$ 11.76	9.84 $\pm$ 2.30	59.64
OpenAI-STK	80.80 $\pm$ 23.55	66.40 $\pm$ 17.49	82.08 $\pm$ 21.02	88.80 $\pm$ 13.98	33.76 $\pm$ 11.95	16.16 $\pm$ 3.87	61.33
MPU	100.00 $\pm$ 0.00	87.92 $\pm$ 1.37	89.84 $\pm$ 4.19	99.28 $\pm$ 0.30	55.84 $\pm$ 3.28	85.52 $\pm$ 3.31	86.40
MPU-STK	100.00 $\pm$ 0.00	97.28 $\pm$ 1.74	99.36 $\pm$ 1.28	99.52 $\pm$ 0.59	74.24 $\pm$ 12.40	82.64 $\pm$ 15.34	92.17

Table 8: Cross-domain performance concerning AUROC%/100. The detection model is trained on ChatGPT text, with the source domain as Essay and the target domain as Reuters.

Method	ChatGPT	GPT-4	ChatGPT-turbo	ChatGLM	Dolly	Claude	Avg.
ChatGPT-D	98.83 $\pm$ 0.60	97.39 $\pm$ 1.11	97.67 $\pm$ 1.36	99.42 $\pm$ 0.22	80.18 $\pm$ 1.55	42.99 $\pm$ 7.00	86.08
ChatGPT-STK	99.27 $\pm$ 0.59	97.56 $\pm$ 1.41	98.07 $\pm$ 0.69	99.39 $\pm$ 0.27	84.41 $\pm$ 4.53	63.22 $\pm$ 12.01	90.32
OpenAI-D	99.46 $\pm$ 0.67	99.03 $\pm$ 0.68	99.44 $\pm$ 0.57	99.52 $\pm$ 0.56	96.15 $\pm$ 0.81	82.63 $\pm$ 1.73	96.04
OpenAI-STK	99.54 $\pm$ 0.25	98.93 $\pm$ 0.51	99.44 $\pm$ 0.27	99.45 $\pm$ 0.38	95.38 $\pm$ 0.96	86.72 $\pm$ 2.16	96.57
MPU	100.00 $\pm$ 0.00	99.39 $\pm$ 0.09	99.60 $\pm$ 0.21	99.97 $\pm$ 0.02	95.97 $\pm$ 0.92	99.48 $\pm$ 0.12	99.07
MPU-STK	100.00 $\pm$ 0.00	99.78 $\pm$ 0.10	99.91 $\pm$ 0.19	99.99 $\pm$ 0.01	97.94 $\pm$ 1.06	99.34 $\pm$ 0.44	99.49

Table 9: Performance comparison (TPR@FPR-0.5%/100) between GPT2 and its enhanced version GPT2-STK. The detection model is trained on ChatGPT text.

Dataset	Method	ChatGPT	GPT4	ChatGPT-turbo	ChatGLM	Dolly	Claude	Avg
Essay	GPT2	33.92 $\pm$ 9.22	35.57 $\pm$ 6.34	38.32 $\pm$ 6.99	41.20 $\pm$ 9.49	11.85 $\pm$ 4.28	13.20 $\pm$ 4.71	29.01
	GPT2-STK	34.40 $\pm$ 9.52	38.93 $\pm$ 7.76	38.80 $\pm$ 6.53	44.98 $\pm$ 6.90	12.45 $\pm$ 3.87	13.04 $\pm$ 4.76	30.43
Reuters	GPT2	79.12 $\pm$ 9.14	71.68 $\pm$ 8.05	76.32 $\pm$ 7.83	74.96 $\pm$ 10.57	36.00 $\pm$ 6.44	53.92 $\pm$ 9.44	65.33
	GPT2-STK	80.48 $\pm$ 8.05	74.16 $\pm$ 9.71	79.52 $\pm$ 6.50	78.32 $\pm$ 8.51	38.80 $\pm$ 7.40	56.72 $\pm$ 7.90	68.00
SQuAD1	GPT2	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.12 $\pm$ 0.23	0.12 $\pm$ 0.23	0.12 $\pm$ 0.23	0.06
	GPT2-STK	0.00 $\pm$ 0.00	0.12 $\pm$ 0.24	0.00 $\pm$ 0.00	0.12 $\pm$ 0.23	0.12 $\pm$ 0.23	0.35 $\pm$ 0.46	0.12

Table 10: Performance comparison (TPR@FPR-0.5%/100) against paraphrase attack Polishing. The detection model is trained on non-paraphrased Google-PaLM text.

Method	Google-PaLM	Claude-instant	ChatGPT	Llama-2-70b	Avg.
ChatGPT-D	79.04 $\pm$ 5.12	58.54 $\pm$ 8.08	83.36 $\pm$ 6.67	94.73 $\pm$ 1.08	78.92
ChatGPT-STK	79.04 $\pm$ 4.59	60.78 $\pm$ 7.98	84.52 $\pm$ 6.12	94.79 $\pm$ 1.27	79.78
OpenAI-D	94.99 $\pm$ 0.90	38.63 $\pm$ 1.60	98.15 $\pm$ 0.37	99.45 $\pm$ 0.20	82.81
OpenAI-STK	95.35 $\pm$ 1.10	40.64 $\pm$ 2.56	98.21 $\pm$ 0.61	99.47 $\pm$ 0.22	83.42
MPU	92.61 $\pm$ 1.05	77.21 $\pm$ 1.41	99.54 $\pm$ 0.30	99.91 $\pm$ 0.09	92.32
MPU-STK	92.43 $\pm$ 0.94	78.24 $\pm$ 1.32	99.58 $\pm$ 0.19	99.92 $\pm$ 0.09	92.54

Table 11: Performance comparison (TPR@FPR-0.5%/100) against paraphrase attack Back-translation. The detection model is trained on non-paraphrased Google-PaLM text.

Method	Google-PaLM	Claude-instant	ChatGPT	Llama-2-70b	Avg.
ChatGPT-D	80.77 $\pm$ 5.93	78.18 $\pm$ 6.47	91.81 $\pm$ 5.27	94.77 $\pm$ 2.90	86.38
ChatGPT-STK	81.49 $\pm$ 5.22	79.29 $\pm$ 5.33	93.39 $\pm$ 3.62	95.62 $\pm$ 1.83	87.45
OpenAI-D	99.17 $\pm$ 0.30	98.57 $\pm$ 0.52	99.42 $\pm$ 0.20	99.20 $\pm$ 0.16	99.09
OpenAI-STK	99.23 $\pm$ 0.23	98.73 $\pm$ 0.36	99.45 $\pm$ 0.20	99.39 $\pm$ 0.25	99.20
MPU	80.64 $\pm$ 2.09	76.30 $\pm$ 3.08	97.30 $\pm$ 0.71	97.90 $\pm$ 0.57	88.03
MPU-STK	80.47 $\pm$ 2.34	77.05 $\pm$ 2.50	97.54 $\pm$ 0.69	98.05 $\pm$ 0.40	88.28

Table 12: Performance comparison (TPR@FPR-0.5%/100) between RADAR and its enhanced version RADAR-STK. The detection model is trained on ChatGPT text.

Dataset	Method	ChatGPT	GPT4All	ChatGPT-turbo	ChatGLM	Dolly	Claude	Avg.
Essay	RADAR	96.88 $\pm$ 2.25	96.56 $\pm$ 0.56	92.64 $\pm$ 4.47	98.96 $\pm$ 0.90	65.75 $\pm$ 8.58	58.64 $\pm$ 8.36	84.90
	RADAR-STK	98.16 $\pm$ 1.38	95.82 $\pm$ 1.78	94.64 $\pm$ 3.89	99.12 $\pm$ 0.59	70.39 $\pm$ 8.22	64.96 $\pm$ 7.80	87.18
Reuters	RADAR	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	99.68 $\pm$ 0.30	99.92 $\pm$ 0.16	90.32 $\pm$ 1.85	96.32 $\pm$ 0.30	97.71
	RADAR-STK	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	99.68 $\pm$ 0.30	99.92 $\pm$ 0.16	90.48 $\pm$ 2.44	96.32 $\pm$ 1.61	97.73
SQuAD1	RADAR	27.05 $\pm$ 14.62	30.71 $\pm$ 10.01	34.27 $\pm$ 10.12	19.54 $\pm$ 13.00	31.74 $\pm$ 4.38	29.48 $\pm$ 11.57	28.80
	RADAR-STK	33.18 $\pm$ 11.40	32.26 $\pm$ 8.40	38.25 $\pm$ 5.54	24.16 $\pm$ 9.88	32.33 $\pm$ 1.93	33.53 $\pm$ 11.31	32.28

Table 13: Running time comparison. Note that DetectGPT and Fast-DetectGPT are zero-shot methods.

Method	Training Time			Inference Time		
	Essay	Reuters	SQuAD1	Essay	Reuters	SQuAD1
Log-Likelihood	42.17	56.29	20.43	0.02	0.02	0.02
Rank	68.93	89.60	28.31	0.02	0.02	0.02
Log-Rank	65.20	82.72	26.73	0.02	0.02	0.02
Entropy	36.99	49.25	20.15	0.02	0.02	0.02
GLTR	72.04	81.32	31.69	0.02	0.02	0.02
DetectGPT	-	-	-	1960.20	2269.46	858.28
Fast-DetectGPT	-	-	-	113.69	126.78	39.60
ChatGPT-D	151.72	167.33	118.65	5.79	5.91	2.41
OpenAI-D	156.23	169.09	129.37	5.81	5.99	2.57
MPU	172.83	189.46	141.20	5.83	5.87	2.62
RADAR	326.79	331.63	254.16	26.54	27.03	22.49
ChatGPT-STK	269.85	252.18	139.54	10.75	10.89	5.67
OpenAI-STK	286.79	255.36	148.24	11.17	10.92	6.08
MPU-STK	301.51	289.77	151.92	11.26	11.69	6.37
RADAR-STK	587.16	590.11	530.20	45.38	46.81	40.56